Convex Optimization Exercises - Sheet 4

Exercise 12. The norm approximation problem for a system of linear equation $Ax = b$ is the problem of minimizing $\|Ax - b\|$, for a given norm $\|\cdot\|$. Show that both

(i) minimize $\|Ax - b\|_\infty$

(ii) minimize $\|Ax - b\|_1$

are equivalent to an LP. Note that in (b) we are looking for an LP that is of polynomial size.

Exercise 13. Let $P = \{x \in \mathbb{R}^n \mid a_i^T x \leq b_i,\ i = 1, \ldots, m\}$ be a polytope. The *Chebyshev center* of P is the center of the largest Euclidean ball $B = \{x_c + u \mid \|u\| \leq r\}$ that lies in P. Model the problem of finding the Chebyshev center as a Linear Program.

Exercise 14. Let $C, D \subseteq \mathbb{R}^n$ be convex sets. Show that the following two sets are also convex.

(i) $-C = \{-c \mid c \in C\}$

(ii) $C + D = \{c + d \mid c \in C, d \in D\}$

Exercise 15. Are the following functions convex, concave or neither?

(i) $f(x_1, x_2) = x_1 x_2$

(ii) $f(x_1, x_2) = \log(x_1 + x_2)$, $\mathrm{dom}(f) = \mathbb{R}^2_{++}$

(iii) $f(x_1, x_2) = \sqrt{x_1 x_2}$, $\mathrm{dom}(f) = \mathbb{R}^2_{++}$

Exercise 16. Show that $f(x) = \log\log(x)$ is concave on $\mathrm{dom}(f) = (1, \infty)$. What about $f(x) = \log\log\cdots\log(x)$? Find two convex functions $g, h\colon \mathbb{R} \to \mathbb{R}$, such that $g \circ h$ is not convex. Can you find an example where h is non-decreasing?

Exercise 17. Let $f\colon \mathbb{R}^{n+m} \to \mathbb{R}$ be a convex function, and let $C \subseteq \mathbb{R}^m$ be a convex set. Show that $g(x) = \inf_{y \in C} f(x, y)$ is also convex. (*Hint:* describe the epigraph of g, then observe that projections of convex sets are convex sets.)

As a consequence, the distance function $x \mapsto \mathrm{dist}(x, C) = \inf_{y \in C} \|x - y\|$ is a convex function for every convex set C.